ORIGINAL ARTICLE

Mitapivat versus Placebo for Pyruvate Kinase Deficiency

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ABSTRACT

BACKGROUND

Pyruvate kinase deficiency is a rare, hereditary, chronic condition that is associated with hemolytic anemia. In a phase 2 study, mitapivat, an oral, first-in-class activator of erythrocyte pyruvate kinase, increased the hemoglobin level in patients with pyruvate kinase deficiency.

METHODS

In this global, phase 3, randomized, placebo-controlled trial, we evaluated the efficacy and safety of mitapivat in adults with pyruvate kinase deficiency who were not receiving regular red-cell transfusions. The patients were assigned to receive either mitapivat (5 mg twice daily, with potential escalation to 20 or 50 mg twice daily) or placebo for 24 weeks. The primary end point was a hemoglobin response (an increase from baseline of ≥1.5 g per deciliter in the hemoglobin level) that was sustained at two or more scheduled assessments at weeks 16, 20, and 24. Secondary efficacy end points were the average change from baseline in the hemoglobin level, markers of hemolysis and hematopoiesis, and the change from baseline at week 24 in two pyruvate kinase deficiency–specific patient-reported outcome measures.

RESULTS

Sixteen of the 40 patients (40%) in the mitapivat group had a hemoglobin response, as compared with none of the 40 patients in the placebo group (adjusted difference, 39.3 percentage points; 95% confidence interval, 24.1 to 54.6; two-sided P<0.001). Patients who received mitapivat had a greater response than those who received placebo with respect to each secondary end point, including the average change from baseline in the hemoglobin level. The most common adverse events were nausea (in 7 patients [18%] in the mitapivat group and 9 patients [23%] in the placebo group) and headache (in 6 patients [15%] and 13 patients [33%], respectively). Adverse events of grade 3 or higher occurred in 10 patients (25%) who received mitapivat and 5 patients (13%) who received placebo.

CONCLUSIONS

In patients with pyruvate kinase deficiency, mitapivat significantly increased the hemoglobin level, decreased hemolysis, and improved patient-reported outcomes. No new safety signals were identified in the patients who received mitapivat. (Funded by Agios Pharmaceuticals; ACTIVATE ClinicalTrials.gov number, NCT03548220.)

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This article was last updated on April 21, 2022, at NEJM.org.

N Engl J Med 2022;386:1432-42.
DOI: 10.1056/NEJMoa2116634
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Pyruvate kinase deficiency, a rare hereditary disease that is characterized by chronic hemolytic anemia, results in numerous serious complications and a reduced health-related quality of life.[1-3] The prevalence of this clinically heterogeneous disease is estimated to be between 3.2 and 8.5 cases per million among Western countries.[2] The condition is characterized by mutations in the *PKLR* gene encoding the red-cell pyruvate kinase enzyme, which is critical for maintaining red-cell energy levels and, therefore, the normal red-cell life span. Owing to the lack of mitochondria, red-cell energy production is entirely dependent on the glycolytic pathway for the formation of ATP, a process catalyzed by pyruvate kinase.[4-6]

Defects in pyruvate kinase cause swelling and premature destruction of erythrocytes (i.e., hemolysis), which manifest clinically as anemia, along with serious complications that include gallstones, pulmonary hypertension, thrombotic complications, osteoporosis, and iron overload. These complications may occur regardless of the degree of anemia or whether a patient receives red-cell transfusions.[7] Independent of complications, chronic anemia itself in patients with pyruvate kinase deficiency is often severe, resulting in highly limiting constitutional symptoms and fatigue.[1,3,8-11]

Clinical management of pyruvate kinase deficiency has been limited to supportive therapies only and has included splenectomy and red-cell transfusions (either at regular or intermittent intervals). Both of these therapies are associated with short-term and long-term risks.[12]

Mitapivat, a first-in-class, oral, allosteric activator of red-cell pyruvate kinase, has been approved by the Food and Drug Administration for the treatment of hemolytic anemia in adults with pyruvate kinase deficiency.[6,13] By activating and stabilizing mutant pyruvate kinase, mitapivat targets the underlying enzymatic defect that causes chronic hemolysis and ineffective erythropoiesis in order to restore enzyme activity.[6] In the phase 2 DRIVE-PK study involving 52 adults with pyruvate kinase deficiency who had not regularly received red-cell transfusions, mitapivat resulted in an early increase from baseline in the hemoglobin level (≥1 g per deciliter) in approximately half the patients.[14] A lack of response was noted in 5 patients who were homozygous for the R479H mutation and in 10 patients who had two non-missense mutations, without another missense mutation, in *PKLR*. An improvement in markers of hemolysis was also noted, and mitapivat had an acceptable safety profile during long-term use for up to 42 months.[11,14,15] On the basis of these phase 2 results, we conducted the global, phase 3 ACTIVATE trial to evaluate the efficacy and safety of mitapivat in patients with pyruvate kinase deficiency who had not regularly received red-cell transfusions.

METHODS

TRIAL DESIGN AND OVERSIGHT

We conducted this multicenter, double-blind, randomized, placebo-controlled trial in North America, western Europe, Asia, the Middle East, eastern Europe, and South America. The trial sites are listed in the Supplementary Appendix, available with the full text of this article at NEJM.org. Table S1 in the Supplementary Appendix summarizes the geographic locations of the patients who underwent randomization. The trial was conducted in accordance with the provisions of the Declaration of Helsinki, the International Council for Harmonisation Good Clinical Practice guidelines, and applicable regulatory requirements. The trial protocol, available at NEJM.org, was approved by an institutional review board or independent ethics committee at each participating institution, and all the patients provided written informed consent. An independent data monitoring committee was established to review the trial data periodically and provide safety oversight for the trial patients.

The trial sponsor, Agios Pharmaceuticals, designed the trial and collected and analyzed the data in collaboration with all the authors. The sponsor was involved in interpretation of the data as well as fact-checking of information provided in the manuscript; however, the authors assume ultimate responsibility for the opinions, conclusions, and data interpretation. A professional writer paid by the sponsor assisted the authors in the preparation of the manuscript. All the authors contributed to the drafting, review, and revision of the manuscript and the decision to submit it for publication. The authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol.

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PATIENTS

Patients who were 18 years of age or older were eligible for enrollment in the trial if they had clinical laboratory confirmation of pyruvate kinase deficiency (defined as the documented presence of at least two mutant alleles in *PKLR*, of which at least one was a missense mutation), a baseline hemoglobin level of 10.0 g per deciliter or less, and adequate organ function. In addition, eligible patients had not regularly received red-cell transfusions (≤4 transfusion episodes in the previous year and no transfusion episodes within 3 months before randomization).

The phase 2 study, DRIVE-PK,[14] showed a lack of response to mitapivat in patients who were homozygous for the R479H mutation or had two non-missense mutations, without another missense mutation, in *PKLR*; thus, such patients were excluded from this trial. Patients who had undergone splenectomy within 12 months before trial enrollment or in whom one was planned, or who had undergone previous bone marrow or stem-cell transplantation, were also excluded. Full eligibility criteria are listed in the trial protocol.

TRIAL PROCEDURES AND TREATMENT

Randomization was stratified according to the mean screening hemoglobin values (<8.5 or ≥8.5 g per deciliter) and the *PKLR* mutation category (missense/missense or missense/non-missense). Patients were stratified and were randomly assigned in a 1:1 ratio with the use of an interactive response system to receive oral mitapivat or placebo. The identical appearance and labeling of the placebo and active-drug tablets ensured that the patients, investigators, and site personnel were unaware of the trial-group assignments.

The patients received mitapivat or placebo for 24 weeks; the trial included a 12-week dose-escalation period and a 12-week fixed-dose period. During the dose-escalation period, mitapivat or placebo was administered at an initial oral dose of 5 mg twice daily, with two potential sequential steps to increase the dose (i.e., from 5 to 20 mg twice daily and from 20 to 50 mg twice daily) on the basis of safety and efficacy assessments. The dose was increased to the next level at week 4 and week 8 of the dose-escalation period if patients had received mitapivat or placebo without unacceptable side effects and the hemoglobin level was less than 2.5 g per deciliter below the upper limit of the normal range for sex. The dose that patients were receiving at the week 12 visit was maintained during the subsequent 12-week fixed-dose period. All the patients who continued to receive mitapivat or placebo through week 24 were eligible to participate in an open-label extension trial in which all the patients received mitapivat.

END POINTS

The primary end point of the trial was a hemoglobin response (an increase from baseline of at least 1.5 g per deciliter in the hemoglobin level) that was sustained at two or more scheduled assessments at weeks 16, 20, and 24 during the fixed-dose period. The key secondary end point was the average change from baseline in the hemoglobin level at weeks 16, 20, and 24. Additional secondary end points included the average change from baseline at weeks 16, 20, and 24 in markers of hemolysis (the indirect bilirubin, lactate dehydrogenase [LDH], and haptoglobin levels) and hematopoietic activity (the reticulocyte percentage), as well as the change from baseline at week 24 in the scores of two patient-reported outcome measures, the pyruvate kinase deficiency impact assessment and the pyruvate kinase deficiency diary[16-18]; these measures are described in the Supplementary Appendix.

Safety assessments included the type, incidence, and severity of adverse events that occurred during the treatment period, as well as the results of clinical laboratory tests and physical examination findings. All adverse events were graded with the use of the Common Terminology Criteria for Adverse Events, version 4.03.[19]

STATISTICAL ANALYSIS

Efficacy analyses were performed in the full analysis set, which consisted of all the patients who had undergone randomization. For the primary end point, we estimated that 76 patients (38 per group), on the basis of an assumed hemoglobin response rate of 35% among patients who received mitapivat and 5% among those who received placebo, would provide the trial with 90% power to detect a treatment effect,

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with the use of a two-sided Fisher's exact test at the 5% significance level. The statistical hypotheses for the primary and secondary end points are described in the Supplementary Appendix.

The primary end point (hemoglobin response) was analyzed with the use of a logistic-regression model that included the trial group as an independent variable and was adjusted for randomization stratification factors. If the logistic-regression model failed to converge or if its maximum likelihood estimate did not exist owing to quasi-complete separation (e.g., if none of the patients in the placebo group had a hemoglobin response), the nonparametric Cochran–Mantel–Haenszel test was used for the primary analysis. Patients who had missing hemoglobin assessments over weeks 16, 20, or 24 and who did not have at least two assessments showing an increase from baseline of at least 1.5 g per deciliter in the hemoglobin level were considered not to have had a response.

For the key secondary end point (the average change from baseline in the hemoglobin level at weeks 16, 20, and 24), estimated between-group differences (least-squares means, 95% confidence intervals, and two-sided P values) were based on the mixed-effects model with repeated measures. The model included the following: the change from baseline in the hemoglobin level as the dependent variable; the baseline hemoglobin level as a covariate; trial group, visit, and trial group–by–visit interaction as fixed effects; and patient as the random effect, with adjustment for stratification factors.

Additional secondary end points, including markers of hemolysis and hematopoietic activities, as well as patient-reported outcome scores, were also assessed with the use of the mixed-effects model with repeated measures. For these end points, a method that was similar to the one described for the key secondary end point was used.

The effect of mitapivat on the primary end point and the key secondary end point across prespecified patient subgroup categories was also examined. Additional details are provided in the Statistical Methodology for Subgroup Analyses section in the Supplementary Appendix.

Safety outcomes were summarized descriptively. The safety analysis set included all the patients who received at least one dose of mitapivat or placebo.

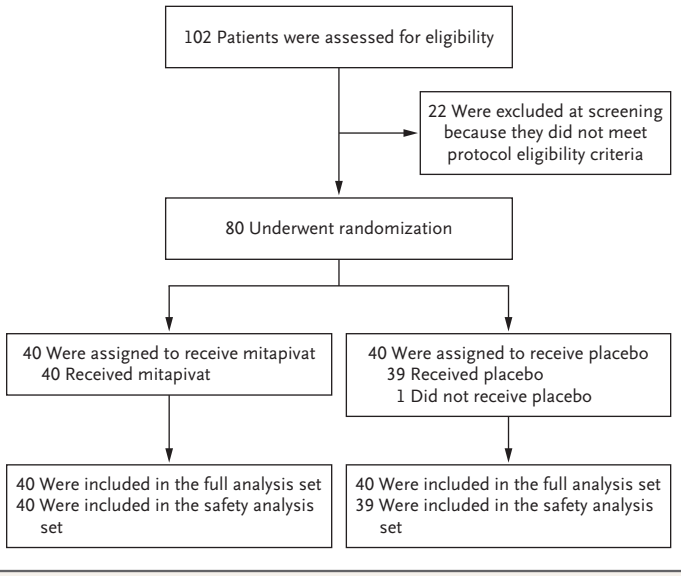


Figure 1. Screening and Randomization of the Patients.

In the placebo group, 1 patient withdrew from the trial before receiving the first dose of placebo; thus, 39 of 40 patients who were randomly assigned to receive placebo were included in the safety analysis set. The full analysis set included all the patients who underwent randomization, and the safety analysis set included all the patients who received at least one dose of mitapivat or placebo.

RESULTS

PATIENT CHARACTERISTICS

Of the 102 patients who had undergone screening, 80 patients met the eligibility criteria and were randomly assigned to receive either mitapivat (40 patients) or placebo (40 patients) (Fig. 1 and Table S2). All 40 patients in the mitapivat group and 39 of 40 patients in the placebo group completed the 24-week trial period; 1 patient withdrew from the trial before receiving the first dose of placebo. The numbers of patients assigned to each escalated dose are summarized in Table S3. The first patient was enrolled on October 1, 2018, and the last patient completed the trial on October 9, 2020.

The demographic and baseline clinical characteristics of the patients were balanced between the mitapivat and placebo groups (Table 1). The mean (±SD) hemoglobin level was 8.6±1.0 g per

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Table 1. Demographic and Clinical Characteristics of the Patients at Baseline in the Full Analysis Set.*

Characteristic	Mitapivat (N=40)	Placebo (N=40)
Mean age (range) — yr	36 (18–70)	37 (19–78)
Sex — no. (%)		
Male	16 (40)	16 (40)
Female	24 (60)	24 (60)
Hemoglobin level — g/dl	8.6±1.0	8.5±0.8
Ferritin level — μg/liter	747.9±1116.2	688.0±605.2
Indirect bilirubin level — μmol/liter	81.8±61.3	89.1±61.8
LDH level — U/liter	348±276.0	260±140.2
Haptoglobin level — g/liter	0.08±0.1	0.08±0.1
Reticulocytes — %	37.1±24.1	40.1±22.2
Previous treatment — no. (%)		
Red-cell transfusions†		
0	29 (72)	30 (75)
1	8 (20)	7 (18)
2	0	1 (2)
3	3 (8)	1 (2)
≥4	0	1 (2)
Splenectomy	28 (70)	30 (75)
Cholecystectomy	28 (70)	30 (75)
Chelation therapy‡	5 (12)	10 (25)
PKLR mutation category		
Missense/missense	28 (70)	27 (68)
Missense/non-missense	12 (30)	13 (32)

* Plus–minus values are means ±SD. The full analysis set included all the patients who underwent randomization. Percentages may not total 100 because of rounding. LDH denotes lactate dehydrogenase, and *PKLR* the gene encoding the pyruvate kinase liver and red-cell isozymes.
† Transfusions within 12 months before the first day of trial treatment are shown.
‡ Patients were considered to have received chelation therapy if they had received chelation therapy within 52 weeks (364 days) before the first dose of mitapivat or placebo.

deciliter in the mitapivat group and 8.5±0.8 g per deciliter in the placebo group. At least 70% of the patients in both groups had undergone previous splenectomy, cholecystectomy, or both. More patients in the placebo group than in the mitapivat group had received chelation therapy in the previous 12 months (25% vs. 12%); patients in both groups had elevated ferritin levels, and more than 50% of the patients had decreased bone mineral density, as indicated by a T score of less than –1.0 (Table S4).

EFFICACY

Primary End Point

Of the 40 patients in the mitapivat group, 16 patients had an increase from baseline of at least 1.5 g per deciliter in the hemoglobin level. This response was sustained at two or more scheduled assessments at weeks 16, 20, and 24 during the fixed-dose period; in contrast, none of the 40 patients in the placebo group had a hemoglobin response. The percentage of patients with a hemoglobin response was significantly higher in the mitapivat group than in the placebo group (40% vs. 0%; adjusted difference, 39.3 percentage points [95% confidence interval {CI}, 24.1 to 54.6]; two-sided P<0.001 by the Cochran–Mantel–Haenszel test) (Fig. 2A). Three patients (8%) in the mitapivat group and 5 patients (12%) in the placebo group were designated as not having had a hemoglobin response because of missing hemoglobin assessments.

Of the 16 patients who had a hemoglobin response in the mitapivat group, the mean (±SD) of the average change from baseline in the hemoglobin level across weeks 16, 20, and 24 was 3.5±1.4 g per deciliter. In addition, all prespecified patient subgroups (including those defined according to a baseline hemoglobin level of <8.5 or ≥8.5 g per deciliter and those defined according to previous splenectomy status) in the mitapivat group had consistent increases in the hemoglobin level, as compared with those in the placebo group (Table S5 and Fig. S2A).

Secondary End Points

Results for all the secondary end points favored mitapivat over placebo (Table 2 and Fig. S3). With respect to the key secondary end point (the average change from baseline in the hemoglobin level at weeks 16, 20, and 24), patients who received mitapivat had a significantly greater response than those who received placebo (least-squares mean change in the hemoglobin level from baseline, 1.7 g per deciliter [95% CI, 1.3 to 2.1] vs. –0.1 g per deciliter [95% CI, –0.6 to 0.3]; least-squares mean difference, 1.8 g per deciliter [95% CI, 1.2 to 2.4]; two-sided P<0.001). This response was both early and sustained (Fig. 2B). In all prespecified patient subgroups, patients who received mitapivat had a consistently greater response with respect to this key secondary end point than those who received placebo (Fig. S2B).

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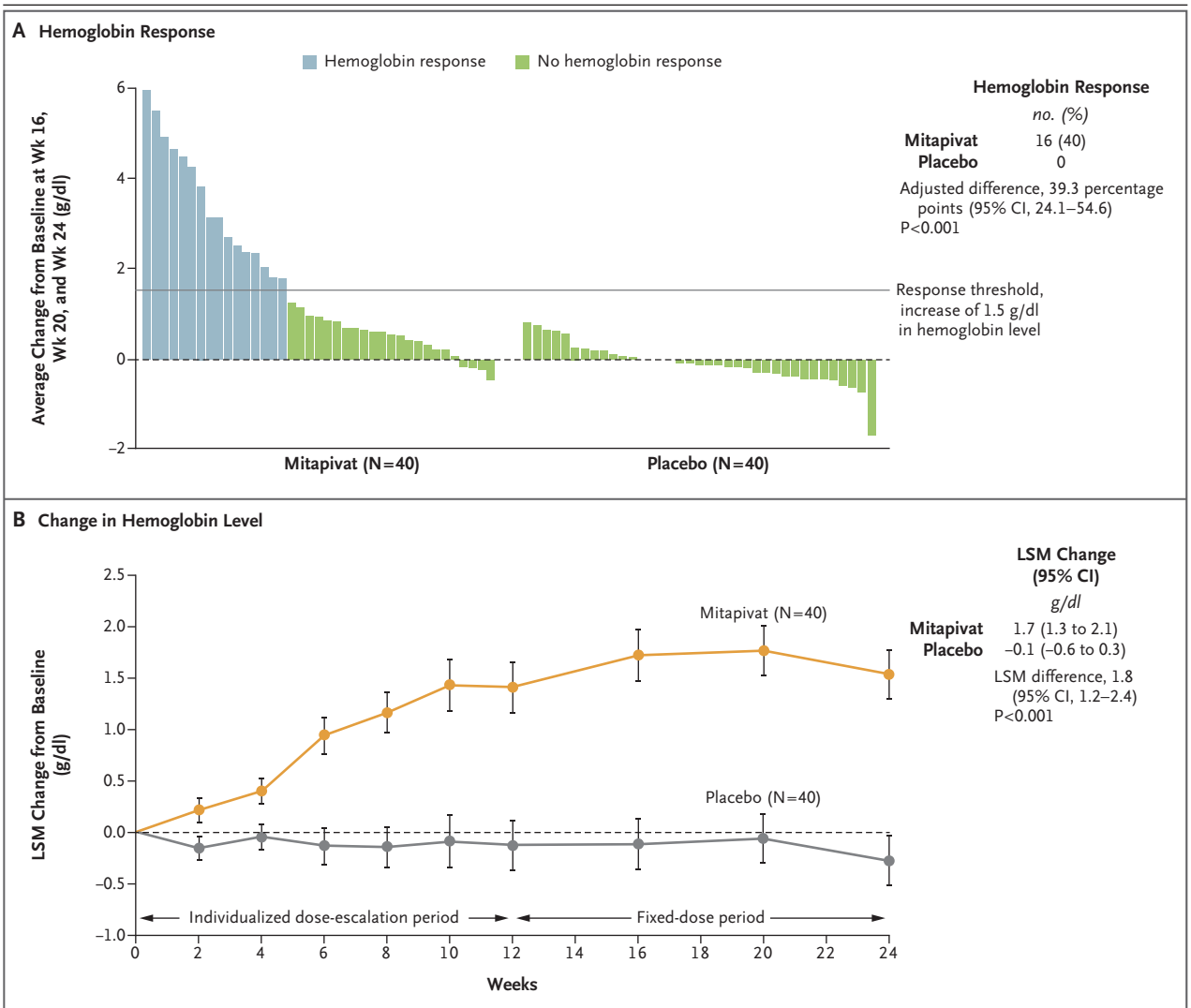


Figure 2. Changes from Baseline in the Hemoglobin Level.

In both panels, baseline was defined as the average of all screening assessments within 45 days before randomization in patients who underwent randomization and did not receive mitapivat or placebo, or before the start of the trial treatment period in patients who underwent randomization and received mitapivat or placebo. Assessments performed within 61 days after a red-cell transfusion were excluded from the baseline derivation. Panel A shows the primary end point of hemoglobin response (defined as an increase from baseline of ≥1.5 g per deciliter in the hemoglobin level) that was sustained at two or more scheduled assessments at weeks 16, 20, and 24 during the fixed-dose period. Each bar represents an individual patient in the full analysis set (all the patients who underwent randomization) who was assigned to receive either mitapivat or placebo. Panel B shows the least-squares mean (LSM) change from baseline in the hemoglobin level in the two groups during the trial period. The I bars indicate the standard error.

Patients who received mitapivat also had a greater response than those who received placebo with respect to the additional secondary end points, including the average change from baseline in markers of hemolysis (the indirect bilirubin, LDH, and haptoglobin levels) and hematopoietic activity (the reticulocyte percentage). The use of mitapivat led to early and sustained improvements in these markers. Improvements from baseline at week 24 in patient-reported outcomes were also greater in the mitapivat group than in the placebo group.

Individual patient profiles, including each patient's genotype and the change in hemoglo-

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Table 2. Least-Squares Mean Changes from Baseline in the Secondary End Points.*

Variable	Mitapivat (N=40)	Placebo (N=40)	Difference (95% CI)	P Value
	LSM Change (95% CI)	LSM Change (95% CI)		
Hemoglobin level — g/dl	1.7 (1.3 to 2.1)	–0.1 (–0.6 to 0.3)	1.8 (1.2 to 2.4)	<0.001
Indirect bilirubin level — μmol/liter	–21.2 (–29.6 to –12.7)	5.1 (–3.0 to 13.2)	–26.3 (–37.8 to –14.7)	<0.001
LDH level — U/liter	–92.0 (–124.5 to –59.5)	–21.2 (–53.3 to 10.9)	–70.8 (–115.9 to –25.7)	0.003
Haptoglobin level — g/liter	0.17 (0.09 to 0.25)	0.01 (–0.07 to 0.09)	0.16 (0.04 to 0.27)	0.008
Reticulocytes — %	–9.7 (–12.5 to –6.9)	0.4 (–2.4 to 3.1)	–10.1 (–13.9 to –6.3)	<0.001
PKDD score†	–5.2 (–7.1 to –3.3)	–2.1 (–4.0 to –0.1)	–3.1 (–5.8 to –0.4)	0.02
PKDIA score‡	–4.7 (–6.9 to –2.4)	–1.4 (–3.7 to 0.9)	–3.3 (–6.4 to –0.1)	0.04

* Shown are the least-squares mean (LSM) of the average change from baseline at weeks 16, 20, and 24 in the hemoglobin level, markers of hemolysis, and hematopoietic activity, and the LSM changes from baseline at week 24 in the patient-reported outcome measures. Baseline for the laboratory variables is defined as the average of all screening assessments within 45 days before randomization in patients who underwent randomization and did not receive mitapivat or placebo, or before the start of trial treatment in patients who underwent randomization and did receive mitapivat or placebo. Assessments performed within 61 days after a red-cell transfusion were excluded from the baseline derivation.

† Baseline for the weekly mean pyruvate kinase deficiency diary (PKDD) score is defined as the average of daily scores collected within 7 days before randomization in patients who underwent randomization and did not receive mitapivat or placebo, or before the start of trial treatment in patients who underwent randomization and did receive mitapivat or placebo. Scores on the PKDD range from 25 to 76, with higher scores indicating a greater disease burden.

‡ Baseline for the pyruvate kinase deficiency impact assessment (PKDIA) score is defined as the last complete assessment (in which the entire questionnaire was completed with no missing items) before randomization in patients who underwent randomization and did not receive mitapivat or placebo, or before the start of trial treatment in patients who underwent randomization and did receive mitapivat or placebo. Scores on the PKDIA range from 30 to 76, with higher scores indicating a greater disease burden.

bin levels and markers of hemolysis and hematopoietic activity over time, grouped according to baseline hemoglobin level and previous splenectomy status, are included in the Patient Profiles sections in the Supplementary Appendix. The hemoglobin response and the change from baseline according to genotype in each individual patient are also summarized in Table S6.

SAFETY

Adverse events were reported in 35 of 40 patients (88%) who received mitapivat and in 35 of 39 patients (90%) who received placebo during the trial period. The most common adverse events in the mitapivat group were nausea (in 7 patients [18%]) and headache (in 6 patients [15%]); the incidences of these events were similar to or lower than those in the placebo group (in 9 patients [23%] and 13 patients [33%], respectively). Adverse events of grade 3 or higher occurred in 10 patients (25%) in the mitapivat group and in 5 patients (13%) in the placebo group; the most common adverse events of grade 3 or higher in the mitapivat group were hypertriglyceridemia and hypertension (both occurring in 2 patients [5%]).

Four patients in the mitapivat group had serious adverse events (gastroenteritis, atrial fibrillation, rib fracture, and musculoskeletal pain, in one patient each). Two patients in the placebo group had serious adverse events (metapneumovirus infection and obstructive pancreatitis, in one patient each). In the mitapivat group, no adverse events led to death or to discontinuation, interruption, or reduction of the dose of mitapivat; two patients who received placebo had a dose interruption owing to an adverse event (Table 3).

DISCUSSION

In this phase 3, placebo-controlled trial involving patients with pyruvate kinase deficiency who had not received regular red-cell transfusions, 40% of those who received mitapivat had a sustained, early, clinically significant hemoglobin

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response. The limited duration of treatment in this trial made it impossible to assess whether the responses would be durable; however, the results hold promise that chronic, lifelong hemolytic anemia in patients with pyruvate kinase deficiency, and its consequent negative effect on health-related quality of life, may be averted in patients with pyruvate kinase deficiency of certain genotypes.[1,11] This finding contrasted markedly with the finding that none of the patients in the placebo group had a hemoglobin response. The patients who received mitapivat also had corresponding early and meaningful improvement in key markers of hemolysis (the indirect bilirubin, LDH, and haptoglobin levels) and hematopoietic activity (the reticulocyte percentage).

The baseline clinical characteristics of the patients enrolled in this trial were representative of those in a real-world population of patients with pyruvate kinase deficiency who do not regularly receive red-cell transfusions; these characteristics indicate the burden of long-term complications of hemolysis.[10,11,20] There was a wide range of hemoglobin values and elevated ferritin levels in the trial patients (despite the lack of regular red-cell transfusions), as well as a high incidence of splenectomy, cholecystectomy, and decreased bone mineral density.

There were some differences between patients with a response and those without a response with respect to the distribution of baseline clinical characteristics, and some patients may have been more likely to have a response to mitapivat than others. However, on the basis of available data, it is not possible to accurately predict which patients will have a response to mitapivat, because responses to mitapivat were observed in all the subgroups defined according to the previously described baseline clinical characteristics.

In addition, the genetic profiles of the patients in this trial varied widely, and most patients had compound heterozygous *PKLR* mutations, with multiple alleles represented only once. Therefore, a large number of unique combinations of alleles were included. However, among the limited number of patients with identical alleles (e.g., patients who were homozygous for p.R510Q), there were patients with a response as well as those without a response; these findings underscore that a response to treatment cannot be predicted on the basis of genotype alone. Previous data have indicated a low likelihood of response in patients with two non-missense mutations or homozygosity for the R479H mutation; patients with these characteristics were not eligible for this trial. However, certain non-missense mutations may have only a minor effect on protein functionality and structure, and processes beyond the mutations themselves (e.g., post-translational and epigenetic modifications) may improve or diminish the likelihood of response.[14,21]

These results confirm that mitapivat, through its novel mechanism of up-regulating the enzymatic activity of pyruvate kinase and thus increasing ATP levels in red cells,[6] successfully addressed the underlying cause of chronic hemolysis and hemolytic anemia in some patients with pyruvate kinase deficiency. Improvements in the pyruvate kinase deficiency–specific patient-reported outcome measures further support the clinical efficacy of mitapivat and its benefit with respect to health-related quality of life.[16,22] Patients who received mitapivat had substantially greater decreases in symptom severity and disease effects at 24 weeks than those who received placebo. Longer follow-up is warranted to assess the durability of these improvements.

The safety profile of mitapivat was consistent with that in a previous study.[14] No new safety signals were identified.

A limitation of our trial is that we were unable to include all the genetic variants that are responsible for pyruvate kinase deficiency, of which more than 300 have been identified.[19] Also, it is not possible to rule out the occurrence of rare, severe adverse events in this trial; this is a standard limitation of clinical trials.

This phase 3, randomized, placebo-controlled trial involving patients with pyruvate kinase deficiency showed that mitapivat significantly increased hemoglobin levels and improved hemolysis, hematopoiesis, and measures of health-related quality of life in a substantial minority of this patient population by specifically targeting the underlying enzymatic defect. The dose of mitapivat was not discontinued, interrupted, or reduced in patients who received this agent. Mitapivat appeared to be a disease-modifying

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Table 3. Adverse Events.*

Adverse Event	Mitapivat (N=40)	Placebo (N=39)
	number of patients (percent)	
Any adverse event	35 (88)	35 (90)
Treatment-related adverse events	23 (58)	14 (36)
Grade ≥3 adverse events	10 (25)	5 (13)
Grade ≥3 treatment-related adverse events†	3 (8)	0
Serious adverse events‡	4 (10)	2 (5)
Adverse events leading to dose reduction of mitapivat or placebo	0	0
Adverse events leading to interruption of mitapivat or placebo	0	2 (5)
Adverse events leading to discontinuation of mitapivat or placebo	0	0
Adverse events leading to death	0	0
Most common adverse events		
All events		
Any grade	35 (88)	35 (90)
Grade ≥3	10 (25)	5 (13)
Nausea		
Any grade	7 (18)	9 (23)
Grade ≥3	0	0
Headache		
Any grade	6 (15)	13 (33)
Grade ≥3	0	0
Nasopharyngitis		
Any grade	5 (12)	6 (15)
Grade ≥3	0	0
Fatigue		
Any grade	5 (12)	4 (10)
Grade ≥3	0	0
Back pain		
Any grade	5 (12)	3 (8)
Grade ≥3	0	0
Diarrhea		
Any grade	4 (10)	7 (18)
Grade ≥3	0	0
Dizziness		
Any grade	4 (10)	3 (8)
Grade ≥3	0	0
Abdominal pain		
Any grade	4 (10)	2 (5)
Grade ≥3	0	0

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Table 3. (Continued.)

Adverse Event	Mitapivat (N=40)	Placebo (N=39)
	number of patients (percent)	
Arthralgia		
Any grade	4 (10)	2 (5)
Grade ≥3	0	0
Hypertriglyceridemia		
Any grade	3 (8)	0
Grade ≥3	2 (5)	0
Dyspnea		
Any grade	3 (8)	4 (10)
Grade ≥3	0	0
Hypertension§		
Any grade	2 (5)	0
Grade ≥3	2 (5)	0
Increased alanine aminotransferase level		
Any grade	1 (2)	6 (15)
Grade ≥3	0	0
Initial insomnia¶		
Any grade	1 (2)	4 (10)
Grade ≥3	0	0
Upper respiratory tract infection		
Any grade	0	4 (10)
Grade ≥3	0	0
Anemia‖		
Any grade	0	2 (5)
Grade ≥3	0	2 (5)

* Listed are adverse events that occurred in at least 10% of the patients in either of the trial groups and adverse events of grade 3 or higher that occurred in at least 5% of the patients in either of the trial groups. Adverse events are presented according to the preferred terms in the *Medical Dictionary for Regulatory Activities*, version 23.1.
† Treatment-related adverse events of grade 3 or higher in the mitapivat group were musculoskeletal pain (in 1 patient) and hypertriglyceridemia (in 2 patients).
‡ Serious adverse events assessed by the investigator are listed. Serious adverse events in the mitapivat group were gastroenteritis, atrial fibrillation, rib fracture, and musculoskeletal pain (in 1 patient each); of the serious adverse events in the mitapivat group, only musculoskeletal pain was considered by the investigator to be related to mitapivat. Serious adverse events in the placebo group were metapneumovirus infection and obstructive pancreatitis (in 1 patient each).
§ One of the patients who had hypertension had an ongoing medical history of hypertension, and the other patient had a risk factor for hypertension (sleep apnea); in both cases, blood-pressure measurements showed no troubling trends.
¶ Initial insomnia was defined as difficulty in falling asleep.
‖ This category includes hemolytic anemia; no cases of acute hemolysis were reported in the context of rapid withdrawal of mitapivat or placebo during this trial.

pharmacotherapy for the management of pyruvate kinase deficiency.

Supported by Agios Pharmaceuticals.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.
A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.
We thank all the patients, their families, and the clinical support staff involved in this trial for their contributions; Fraser Macleod, M.B., Ch.B., for medical writing support; and Elke Sims, M.Lang.Trans., of Onyx Medica, London, for editorial support with an earlier version of the manuscript.

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APPENDIX

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